CB-SAFE: Code-Based Post-Quantum Secure Aggregation for Byzantine-Robust Federated Learning

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***Abstract*—Secure aggregation protects federated learning (FL) updates by revealing only their sum, but its key establishment breaks under a quantum adversary, and post-quantum FL today derives confidentiality almost exclusively from lattice assumptions, a single point of cryptanalytic failure. We present CB-SAFE, to our knowledge the first FL secure-aggregation framework whose post-quantum confidentiality is established by a code-based KEM (HQC, selected by NIST in March 2025 to diversify away from lattices) rather than a lattice KEM. CB-SAFE treats the KEM as a pluggable module: the same double-masking protocol runs over HQC or ML-KEM, and per-round mask seeds are derived from long-lived encapsulated secrets, so the KEM affects only a one-time setup. Measured on a 291 498-parameter model with 30 clients, per-round traffic is KEM-independent (2 277 KiB per client), and the entire code-based premium is a one-time 15.2 vs. 3.8 KiB per client at NIST level 1 (49.4 vs. 7.7 KiB at level 5),** 0.7% **of one round's traffic and under** 0.03% **amortized over training. CB-SAFE further reconciles secure aggregation with Byzantine robustness by aggregating securely *within* clusters of** c **clients and applying robust rules (trimmed mean, median, Multi-Krum) *across* the revealed cluster sums, and we quantify the resulting privacy–robustness tension: the anonymity set grows with** c **while the probability a cluster sum stays uncontaminated decays as** $(1-f)^c$ **under malicious fraction** f**. We prove a simulation-based privacy theorem, identify a *laundering* effect by which cluster averaging turns amplified sign-flip poisoning into plausible-magnitude inverted sums that defeat all coordinate-wise robust rules, and introduce CB-SAFE+: re-randomized clustering with root-anchored direction flags accumulates per-client suspicion across rounds (temporal group testing over private sums) and excludes attackers the server never observes individually, revealing no update content beyond the cluster sums CB-SAFE already exposes. Experiments across four datasets in two modalities (non-IID CIFAR-10, FashionMNIST, and EMNIST, and tabular Edge-IIoTset IoT intrusion detection) with label-flipping, sign-flipping, and backdoor attacks validate the framework: the laundering effect and CB-SAFE+'s recovery replicate on every dataset (CB-SAFE+ beats every baseline under sign-flip, Holm-corrected** $p<0.01$**), and the implementation is released open-source.**

***Index Terms*—Post-quantum cryptography, federated learning, secure aggregation, HQC, code-based cryptography, Byzantine robustness, cryptographic diversity.**

I. Introduction

FEDERATED learning (FL) trains a shared model across many clients without centralizing data [24], [25], but the transmitted model updates leak private information: gradient-inversion [26], [27] and membership- or model-inversion [28], [29] attacks reconstruct training samples from individual updates. Secure aggregation closes this channel by ensuring the server observes only the *sum* of updates, using pairwise additive masks that cancel on aggregation [1], [12]; recent designs reduce its overhead [31], [32], though the sum itself can still leak [30]. The confidentiality of those masks rests on key establishment that Shor's algorithm [33] breaks once a cryptographically relevant quantum computer exists, and traffic recorded today can be decrypted then ("harvest-now, decrypt-later").

The standard response replaces classical key exchange with a post-quantum KEM, and nearly all post-quantum FL does so with *lattice* cryptography [4]–[7], [17]. Lattices are efficient and standardized (ML-KEM, FIPS 203 [3]), but concentrating an entire ecosystem's confidentiality on one hardness assumption is a systemic risk: a single cryptanalytic advance would break every such deployment simultaneously. NIST recognized exactly this risk when, in March 2025, it *selected* the code-based KEM HQC (whose security reduces to syndrome decoding of quasi-cyclic codes [35], not to lattice problems, in the tradition of code-based cryptography since McEliece [34]) as a diversified backup to ML-KEM [2]. That diversity has not reached federated learning.

A second, orthogonal tension is that secure aggregation and Byzantine robustness pull in opposite directions: hiding individual updates denies the server exactly the visibility that poisoning defenses need. Recent work combines robustness with *lattice*-based post-quantum aggregation [7], or achieves cluster-based robust aggregation on *quantum hardware* [8]; no prior framework provides both under a code-based, classical post-quantum assumption.

This paper presents CB-SAFE (Code-Based Secure Aggregation for Federated lEarning), making five contributions:

1) **A KEM-agnostic secure-aggregation protocol** (Definition 1) that formalizes Bonawitz-style double masking over any IND-CCA2 KEM, with per-round mask seeds derived from long-lived encapsulated secrets so key encapsulation is a one-time cost. The post-quantum primitive becomes a single replaceable module.
2) **The first code-based instantiation, scope-qualified:** to our knowledge, CB-SAFE is the first FL secure-aggregation framework whose post-quantum confidentiality comes from the NIST-selected code-based KEM (HQC) rather than a lattice KEM, at matched NIST security levels.
3) **A quantified reconciliation of privacy and robustness:** secure aggregation *within* clusters of c clients, robust aggregation (trimmed mean, median, Multi-Krum) *across* cluster sums, and an explicit trade-off law (the anonymity set is c while the clean-cluster probability

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decays as $(1-f)^c$) validated experimentally under three attacks. We further identify and analyze a *laundering* effect: cluster averaging turns amplified sign-flip poisoning into plausible-magnitude, direction-inverted cluster sums that defeat every coordinate-wise robust rule (Proposition 1).

4) **CB-SAFE+, a defense that identifies hidden attackers:** per-round re-randomized clustering plus direction flags against a small server root dataset accumulate per-client suspicion (temporal group testing over privacy-preserving observations) and exclude malicious clients the server has never observed individually. The flags are post-processing of already-revealed cluster sums, so the defense adds *no* update-content leakage beyond CB-SAFE itself (Theorem 1).
5) **An honest, measured cost analysis:** on a 291k-parameter model, per-round traffic is KEM-independent and the full code-based premium is a one-time 15.2 KiB vs. 3.8 KiB per client (NIST level 1), i.e., 0.7% of one round's traffic and under 0.03% amortized over training; HQC's $\sim$200$\times$ slower operations contribute about 2 s (one-time, 30 clients) versus 0.02 s for ML-KEM.

II. Related Work

Post-quantum FL is lattice-dominated. PQSF reconstructs masks with lattice multi-stage secret sharing [4]; Qin and Xu build single-round aggregation from a lattice key-homomorphic PRF [5]; Narsimhulu *et al.* aggregate Dilithium-style lattice signatures [6]; and a 2026 framework combines reputation-weighted Byzantine-robust aggregation with Kyber/lattice-based secure aggregation for IoT threat intelligence [7]. Surveys confirm lattice dominance across post-quantum deployment [17]. Non-lattice exceptions are rare and code-free: multivariate cryptography with blockchain [9], and CQSA, which partitions clients into clusters aggregated via GHZ states on near-term *quantum hardware* [8]. CB-SAFE differs from all of the above on the confidentiality assumption: syndrome decoding of quasi-cyclic codes, via a standard-track classical KEM [2], with the KEM exposed behind an agnostic interface so lattice and code-based variants differ in exactly one module.

Byzantine robustness and poisoning. Model- and data-poisoning attacks [21], [22], [39], including stealthy [37] and backdoor [15], [38] variants, and their defenses are a large literature: robust aggregation rules such as Multi-Krum [13], coordinate-wise trimmed mean and median [14], Bulyan [19], and the geometric-median aggregator RFA [20]; trust-bootstrapped filtering (FLTrust [10], Zeno [11]); malicious-client detection (FLDetector [40]); and backdoor-specific defenses (FLAME [23]). These operate on *individual* client updates, which secure aggregation deliberately hides, creating the privacy–robustness tension we address.

Cluster-based robust aggregation. Reconciling Byzantine resilience with secure aggregation is an established goal: BREA does so through secret-shared distance computation [16], CQSA through clustered aggregation on quantum hardware [8], and, closest to our robustness mechanism, FedGT [18] identifies malicious clients by group testing over overlapping secure-aggregation groups. Aggregating within groups and inspecting group sums is thus a known structure; our contribution is not the cluster topology itself but (i) its first classical post-quantum, code-based instantiation and (ii) an explicit, experimentally validated statement of the privacy–robustness trade-off it induces (Section III-D), with a temporal group-testing defense (CB-SAFE+) that matches FedGT's attacker identification at far lower false-positive cost, and scales past FedGT's fixed small-population design.

TABLE I
Summary of notation.

Symbol	Meaning
N	total number of clients
c	cluster size (anonymity set)
$k = N/c$	number of clusters per round
f	fraction of malicious clients
m	malicious members in a given cluster
d	model dimension (parameters)
α	Dirichlet concentration (label skew)
$\mathbf{x}_i$	local update of client i
$q(\cdot)$	fixed-point quantization (2^{-16}/coord.)
$\mathbf{y}_i$	masked submission of client i
$\mathcal{K}$	KEM $(\mathrm{KeyGen}, \mathrm{Encaps}, \mathrm{Decaps})$
ss_{ij}	pairwise KEM shared secret (i, j)
H, G	key-derivation function, PRG
b_i^r	self-mask seed of i in round r
t	Shamir threshold $\lfloor c/2 \rfloor + 1$
$\mathbf{S}_\ell$	mean of cluster ℓ revealed to server
γ	sign-flip amplification factor
γ^*	laundering boundary $2c/m - 1$
$\mathbf{h}$	honest update direction
$\boldsymbol{\mu}$	contaminated cluster mean
β	trim fraction per side (robust rule)
$\mathcal{D}_0$	server root dataset ($\lvert\mathcal{D}_0\rvert$=200)
$L(\cdot; \mathcal{D}_0)$	root loss probe
g_ℓ^r	flag bit of cluster ℓ in round r
s_i^r	accumulated suspicion of client i
p_h	honest co-cluster flag prob. $1-(1-f)^{c-1}$
τ	exclusion threshold
o	overlap degree (groups per client)
R	number of training rounds
ASR	backdoor attack success rate

III. The CB-SAFE Framework

A. Notation

We use lowercase bold for vectors and uppercase bold for matrices. The server coordinates N clients grouped into clusters of size c; a fraction f is malicious. Table I summarizes the symbols used throughout.

B. System and Threat Model

A server coordinates N clients running FedAvg over non-IID data partitions. Clients are partitioned into $k = N/c$ fixed clusters of size c. The server is honest-but-curious; a fraction f of clients are malicious and submit poisoned updates; clients may drop out mid-round. Within a cluster, mask recovery uses a t-of-c threshold with $t = \lfloor c/2 \rfloor + 1$, so privacy holds against

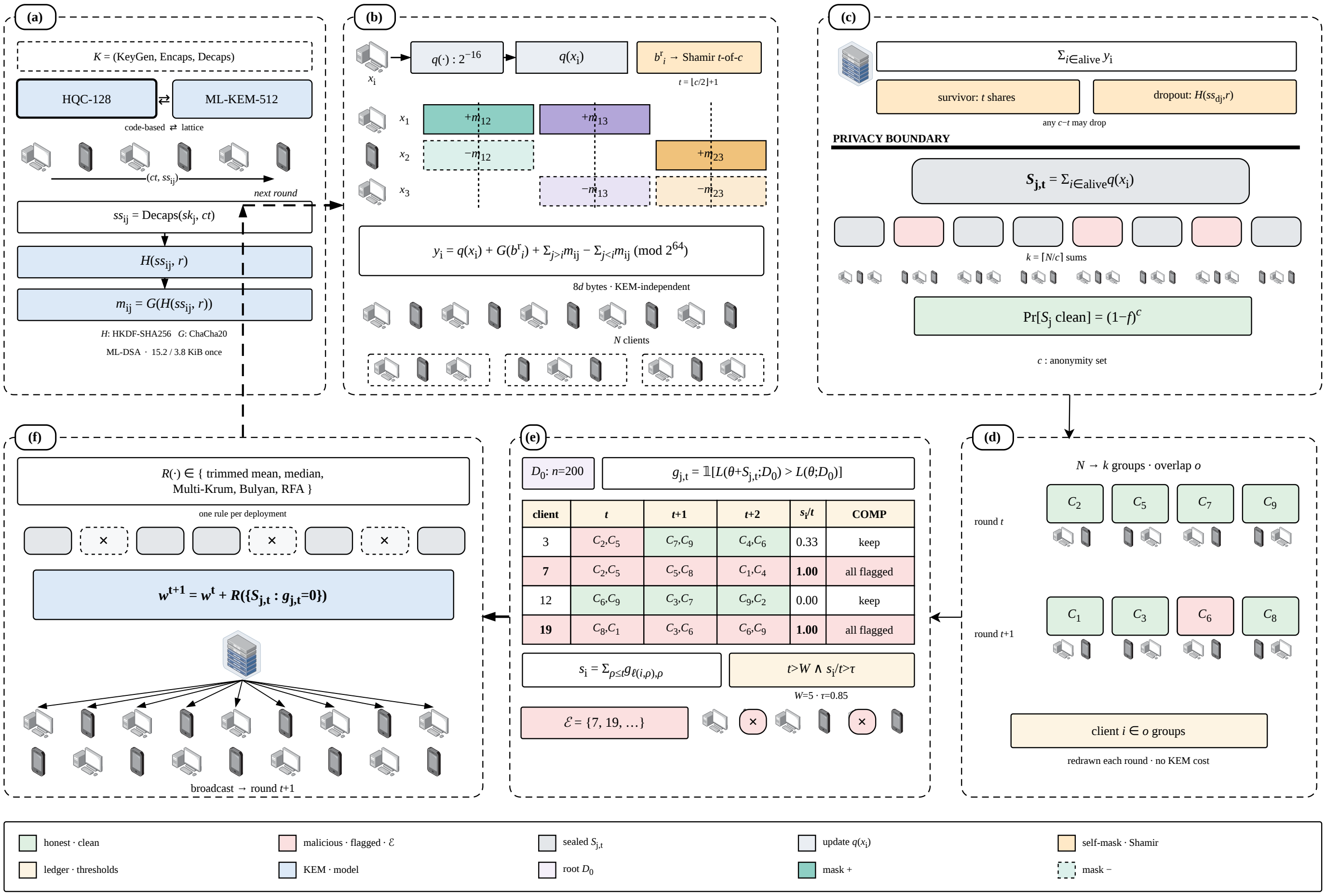

Fig. 1. CB-SAFE, one training round end to end: one-time HQC key setup, within-cluster masked aggregation (the server sees only the k cluster sums), and the CB-SAFE+ flag/exclude stage.

the server colluding with up to $t-1$ members of a cluster, and any $c-t$ members can drop without stalling the round. Update authentication is out of the code-based scope: HQC cannot sign and no code-based signature is standardized, so where signatures are required we compose the lattice-based ML-DSA [36] and state explicitly that only the confidentiality layer is code-based.

C. KEM-Agnostic Masked Aggregation

Definition 1 (KEM-agnostic cluster masking): Let $\mathcal{K} = (\mathrm{KeyGen}, \mathrm{Encaps}, \mathrm{Decaps})$ be an IND-CCA2 KEM, H a key-derivation function, and G a PRG into $(\mathbb{Z}_{2^{64}})^d$. At setup, each client i publishes pk_i; for each cluster pair $i < j$, client i computes $(ct_{ij}, ss_{ij}) = \mathrm{Encaps}(pk_j)$ and j recovers $ss_{ij} = \mathrm{Decaps}(sk_j, ct_{ij})$. In round r, client i quantizes its update x_i to $q(x_i) \in (\mathbb{Z}_{2^{64}})^d$, draws a self-mask seed b_i^r, Shamir-shares it t-of-c within its cluster, and submits

$$\mathbf{y}_i = q(\mathbf{x}_i) + G(b_i^r) + \sum_{j>i} G\big(H(ss_{ij}, r)\big) - \sum_{j<i} G\big(H(ss_{ij}, r)\big) \pmod{2^{64}}, \tag{1}$$

where the sums run over the cluster peers of i. The server adds each cluster's submissions (pairwise masks cancel), reconstructs each survivor's b_i^r from t shares and removes $G(b_i^r)$; for a dropped client d, each survivor j reveals the per-round seed $H(ss_{dj}, r)$ and the server removes the orphaned mask halves. The server thus obtains exactly $\sum_{i \in \mathrm{alive}} q(x_i)$ per cluster.

Correctness is exact: masking is modular-arithmetic exact, and the only deviation from plain float aggregation is fixed-point quantization (2^{-16} per coordinate). Because encapsulation happens once and per-round seeds come from $H(ss_{ij}, r)$, per-round messages contain no KEM material at all: the masked update is $8d$ bytes regardless of the KEM, and the KEM's key and ciphertext sizes appear only in the one-time setup. Instantiating $\mathcal{K}$ with HQC gives the code-based variant; with ML-KEM [3], the lattice baseline.

Theorem 1 (Privacy against honest-but-curious coalitions): Assume $\mathcal{K}$ is IND-CCA2, H is a secure KDF, and G is a secure PRG. Fix a cluster C of size c with threshold $t = \lfloor c/2 \rfloor + 1$, and let the adversary $\mathcal{A}$ comprise the honest-but-curious server and any statically chosen $T \subset C$ with $|T| \le t-1$ (the corruption set is fixed before the protocol runs; we do not claim adaptive-corruption security). There exists a PPT simulator that, given only the corrupted clients' inputs and the sum $\sum_{i \in \mathrm{alive}(C) \setminus T} q(x_i)$, produces a view computationally indistinguishable from $\mathcal{A}$'s view of a CB-SAFE round. The CB-SAFE+ flag bits are a deterministic function of the revealed cluster sums and server-local data, so CB-SAFE+ leaks nothing beyond CB-SAFE.

Algorithm 1: Cluster Secure Aggregation (round r)

Input: cluster C of size c, threshold t, per-pair secrets $\{ss_{ij}\}$
Output: cluster sum $\sum_{i\in\text{alive}} q(\mathbf{x}_i)$
foreach *client* $i \in C$ **do**
  $q(\mathbf{x}_i) \leftarrow$ quantize local update
  draw self-mask seed b_i^r; Shamir t-of-c share it to peers
  send masked $\mathbf{y}_i$ by Eq. (1)
$Y \leftarrow \sum_{i\in\text{alive}} \mathbf{y}_i$ `// pairwise masks cancel`
foreach *survivor* i **do**
  reconstruct b_i^r from t shares; $Y \leftarrow Y - G(b_i^r)$
foreach *dropout* d **do**
  survivors reveal $H(ss_{dj}, r)$; remove orphaned mask halves from Y
return Y

Proof sketch. A hybrid argument. (H1) Replace each pairwise secret ss_{ij} between honest clients by a uniform string: distinguishing H1 from the real game breaks IND-CCA2 of $\mathcal{K}$. (H2) Replace the per-round seeds $H(ss_{ij}, r)$ and mask streams $G(\cdot)$ by uniform: breaks the KDF/PRG. In H2 every honest submission y_i is uniform subject to the alive-honest submissions summing to the revealed cluster sum, which the simulator can sample directly. Shamir shares held by $\leq t-1$ corrupted parties are uniform by perfect secrecy below the threshold, and a dropped client's revealed seeds $H(ss_{dj}, r)$ are independent of other rounds' seeds under the KDF assumption, so both recovery paths are simulatable. Distinct rounds use distinct seeds, so views compose across rounds. □

The anonymity set of an honest client is therefore its cluster's alive honest-member count, and re-randomizing the partition each round (CB-SAFE+) does not change the per-round argument because each round reveals sums of fresh, independently masked updates.

D. Robust Aggregation Across Cluster Sums

The server sees k cluster sums $S_1, \ldots, S_k$ (means after dividing by alive counts) and no individual update. CB-SAFE applies a robust rule across the cluster means: coordinate-wise trimmed mean (trim β per side) or median [14], Multi-Krum [13], Bulyan [19], or the geometric median (RFA) [20]. This resolves the hide-versus-inspect tension structurally (privacy is enforced below the cluster boundary, robustness operates above it) at a quantifiable price. A cluster mean is contaminated if it contains at least one malicious member, so with malicious fraction f and random cluster assignment,

$$\Pr[\text{cluster clean}] = (1-f)^c, \tag{2}$$

and the expected number of clean cluster means is $k(1-f)^c$. Larger c improves privacy (bigger anonymity set) but shrinks the clean fraction the robust rule can rely on; a trimmed mean discarding β per side tolerates at most $\lfloor \beta k \rfloor$ contaminated clusters. c is therefore a tunable privacy–robustness dial, and Section V traces its consequences empirically. Setting c=1 recovers classical (privacy-free) robust aggregation; setting c=N recovers full secure aggregation with no robustness.

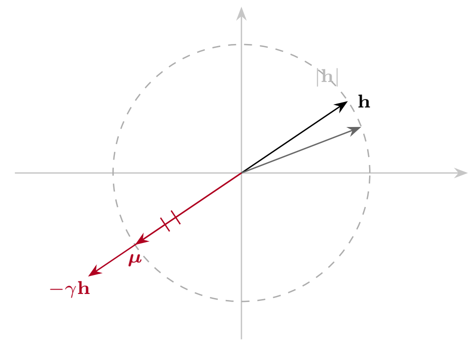

Fig. 2. Laundering geometry (schematic). Averaging $c-m$ honest updates $\mathbf{h}$ with m sign-flips $-\gamma\mathbf{h}$ gives $\boldsymbol{\mu} = \frac{c-m(1+\gamma)}{c}\mathbf{h}$; at γ^*=$2c/m-1$ it matches $|\mathbf{h}|$ with opposite sign (Proposition 1).

The contamination probability, however, understates the problem: cluster averaging also *disguises* what it contaminates.

Proposition 1 (Laundering by cluster averaging): Let each honest update be h (up to noise) and let m members of a cluster of size c submit the scaled sign-flip $-\gamma h$. The cluster mean is $\mu = \frac{c-m(1+\gamma)}{c} h$. Its magnitude equals an honest cluster's ($|\mu| = |h|$) exactly when $\gamma = 2c/m - 1$, while its direction is inverted. At the common attack setting γ=5 with c=3, m=1 this holds with equality: the poisoned cluster sum is indistinguishable from honest by any magnitude- or order-statistic test, and a near-symmetric $\pm h$ population drives coordinate-wise trimmed mean and median toward 0, stalling learning entirely.

This laundering effect explains the empirical collapse of every coordinate-wise rule in Section V and motivates direction-based detection: what cluster averaging cannot hide is the *sign* of the update. The proposition idealizes honest updates as a common h; under non-IID heterogeneity real honest updates vary, so it predicts the boundary γ^* in the mean rather than exactly per cluster. We therefore read the four-dataset collapse as strong empirical validation *near* that boundary, not as a guarantee of collapse for arbitrary honest-update distributions.

E. CB-SAFE+: Temporal Group Testing over Cluster Sums

CB-SAFE+ turns the laundering signature into an identification mechanism, using three ingredients. (i) *Re-randomized clustering*: the partition is redrawn every round (pairwise secrets with all peers are established once at setup, so re-clustering adds no per-round KEM cost). (ii) *Loss-probe flags*: the server maintains a small root dataset ($|\mathcal{D}_0|$=200 samples held out from all clients, the standard trust-bootstrapping assumption [10], [11]), applies each cluster mean to the current model θ^r, and flags every cluster whose mean *increases* root loss,

$$g_\ell^r = \mathbb{1}[\, L(\theta^r + \mathbf{S}_\ell; \mathcal{D}_0) > L(\theta^r; \mathcal{D}_0)\,]\,. \tag{3}$$

Flags are computed from already-revealed cluster sums and server-local data, so they reveal no update content beyond those sums (Theorem 1). The probe is the product of two measured failures. Anchor-free direction flags are sign-ambiguous:

the poisoned cluster sums form their own coherent cloud at $-h$, and a Krum-based reference locks onto that cloud in a large fraction of rounds. Anchored *cosine* flags break the symmetry but lose specificity under non-IID data: an $m{=}1$ poisoned cluster mean is dominated by $-\gamma$ times one client's idiosyncratic direction, nearly orthogonal to the balanced anchor, so its sign test approaches a coin flip (at $f{=}0.2$ we measured cosine flags hitting clean clusters as often as dirty ones, and suspicion never separated). The loss probe is immune to this decorrelation because a γ-amplified ascent direction raises root loss regardless of whose data dominates the cluster sum. (iii) *Suspicion accumulation and exclusion*: every member of a flagged cluster gains one suspicion count, so after r rounds client i has accumulated

$$s_i^r = \sum_{\rho=1}^{r} g^{\rho}_{\ell(i,\rho)}, \tag{4}$$

where $\ell(i,\rho)$ is the cluster containing i in round ρ. A malicious client sits in a flagged cluster in essentially every round ($\mathbb{E}[s_i^r/r] \to 1$), while an honest client is flagged only when randomly co-clustered with an attacker, i.e., with probability

$$p_h = 1 - (1-f)^{c-1} < 1. \tag{5}$$

A client is excluded once its flagged fraction exceeds a threshold $\tau \in (p_h, 1)$ after a warmup of W rounds,

$$\text{exclude } i \iff r > W \text{ and } s_i^r/r > \tau. \tag{6}$$

By Hoeffding's inequality the two populations separate exponentially: the per-client misclassification probability is at most $e^{-2r(\tau-p_h)^2}$, so a few rounds suffice to isolate the attackers, as the measured suspicion trajectories confirm (Fig. 7). The server thereby identifies individual attackers it has never observed individually: group testing [41], [42] where the pools are the privacy-preserving cluster sums and the tests repeat over training rounds. The *hybrid* variant augments Eq. (4) with a per-round combinatorial (COMP) decode over the overlapping groups (degree o), so a client is suspected only if *all* its groups are flagged, further suppressing false positives.

Algorithm 2: CB-SAFE+: Temporal Group Testing over Cluster Sums

```
Input: clients [N], root D_0, cluster size c, threshold τ,
       warmup W, rounds R
Output: global model θ, excluded set ℰ
s_i ← 0 ∀i;   ℰ ← ∅
for r ← 1 to R do
    redraw a random partition of [N] \ ℰ into clusters
      of size c
    foreach cluster ℓ do
        S_ℓ ← ClusterSecureAgg(ℓ, r)/|alive(ℓ)|
          // Alg. 1
        g_ℓ^r ← loss-probe flag, Eq. (3)
        foreach active i ∈ ℓ with g_ℓ^r=1 do s_i ← s_i + 1
    if r > W then
        foreach active i with s_i/r > τ do ℰ ← ℰ ∪ {i}
    θ ← θ + RobustAgg({S_ℓ : g_ℓ^r=0})
return θ, ℰ
```

IV. Implementation

CB-SAFE is implemented in Python on liboqs 0.15.0 (via `liboqs-python`) and PyTorch, with the KEM selected by a one-line configuration change among HQC-128/192/256 and ML-KEM-512/768/1024 (Kyber round-3 parameter sets are retained for comparison with older literature). Masks are ChaCha20 keystreams; per-round seeds use HKDF-SHA256; self-mask seeds are shared with Shamir's scheme over $\mathrm{GF}(2^{127}-1)$. *HQC provenance:* liboqs 0.15.0 ships the PQClean constant-time implementation of the Round-4 HQC specification (2023-04-30), the version NIST evaluated and selected [2]; the finalized standard's parameters may differ slightly once published (a draft is expected as FIPS 207). Federated training is simulated in-process; every protocol byte count and crypto timing reported below is measured from real objects on the wire path of the protocol (public keys, ciphertexts, shares, masked vectors), not modeled. *Reproducibility:* Python 3.12, PyTorch 2.5.0 (CUDA 12.4), liboqs-python 0.15.0, on a single NVIDIA RTX 2060 (6 GB); one training round of 30 clients takes ≈21 s. All runs use fixed seed 0 with deterministic partitioning and cluster assignment; results are averaged over three seeds at 50 rounds, significance is by paired t-test with Holm correction, and every experiment is reproducible from a per-run CSV and the released scripts.

V. Evaluation

A. Setup

Task and data. Primary task: CIFAR-10 partitioned across $N{=}30$ clients by a Dirichlet($\alpha{=}0.5$) label-skew split; a CNN with $d{=}291\,498$ parameters; one local epoch per round (SGD, lr 0.01, momentum 0.9, batch 64); $k{=}10$ clusters of $c{=}3$ unless stated. The split is sharply non-IID: clients hold 631 to 3298 samples with strongly skewed label distributions, the heterogeneity under which all robustness results are reported.

Generalization (Section V-F) adds FashionMNIST (same CNN) and Edge-IIoTset (a 15-class tabular IoT intrusion-detection set with a multilayer perceptron), each Dirichlet-partitioned identically. **Configurations.** Plain FedAvg (no cryptography); CB-SAFE/lattice (ML-KEM); CB-SAFE/code-based (HQC), at matched NIST levels. **Attacks.** Label flipping ($y \to 9-y$), sign flipping (-5Δ), and a backdoor [15] (3×3 trigger patch, target class 0, 50% local poison rate) at malicious fractions $f \in \{0.1, 0.2, 0.3\}$, against aggregation rules mean, trimmed mean, median, Multi-Krum, and the CB-SAFE+ reputation defense (Section III-E; server root set of 200 samples held out from all clients). **Metrics.** Test accuracy; backdoor attack success rate (ASR: fraction of triggered non-target test samples classified as the target); communication bytes per client (setup vs. per-round); crypto wall-time; dropout-recovery cost.

B. Overhead: the Code-Based Premium Is a One-Time Cost

Table II reports measured per-client traffic and crypto time. Per-round traffic is identical across all six KEMs (2 277 KiB

TABLE II
MEASURED PER-CLIENT OVERHEAD (N=30, c=3, d=291 498). SETUP IS ONE-TIME; PER-ROUND TRAFFIC IS KEM-INDEPENDENT.

KEM	Setup (KiB)	Setup wall (s)	Round (KiB)	Mask (ms)
ML-KEM-512	3.8	0.02	2 277	25
HQC-128	15.2	2.0	2 277	26
ML-KEM-768	5.6	0.02	2 277	27
HQC-192	30.8	5.7	2 277	25
ML-KEM-1024	7.7	0.02	2 277	25
HQC-256	49.4	10.6	2 277	26

per client, dominated by the $8d$-byte masked update plus 102 B of Shamir material) because Definition 1 keeps KEM objects out of the per-round path entirely. The KEM appears only in setup: at NIST level 1, HQC costs 15.2 KiB per client against ML-KEM's 3.8 KiB ($4.0\times$; $6.4\times$ at level 5), and 2.0 s of one-time compute for all 30 clients against 0.02 s. Amortized over 40 training rounds ($\sim$89 MiB of per-client update traffic), the code-based premium is below 0.03% of total communication (Table II). A round with 10% dropouts adds only seed-reveal messages (32 B per survivor–dropout pair) and kept server unmasking below 0.28 s in all configurations. The practicality question that motivated this work has a quantitative answer: *cryptographic diversity in FL secure aggregation is essentially free at the protocol level once key establishment is amortized*: the trade-off is a one-time setup constant, not a per-round tax.

C. Scalability in the Client Population

Secure-aggregation cost is set by how many pairwise partners each client must maintain. All-pairs masking [1] gives every client $N-1$ partners, so per-client setup traffic and per-round mask derivation grow as $O(N)$. CB-SAFE masks only *within* clusters of size c, capping partners at $c-1$, so both quantities are *constant in* N. Extrapolating the measured HQC-128 and ML-KEM-512 primitives (Table II) by partner count, at N=1000 a client's one-time setup is 15.2 KiB (HQC) or 3.8 KiB (ML-KEM) regardless of population, versus 7.4 MiB for all-pairs masking, a $500\times$ reduction that widens linearly with N (final column of Table III). Because the privacy law $\Pr[S_j \text{ clean}]=(1-f)^c$ and the laundering boundary (Prop. 1) are stated *per cluster*, they are independent of N, and the re-randomized group tests add no per-round KEM cost (Fig. 1d). CB-SAFE therefore targets the cross-device regime where all-pairs masking is infeasible.

Robustness holds as the population scales. Cost is only half the scaling question; the other is whether robustness survives a $16\times$ larger client population. We ran every rule that admits a configuration at N=500 under sign-flip (f=0.2) for 50 rounds (Table III, Fig. 3). FedGT is necessarily absent here: its parity-check matrices and prevalence-estimation tables are hardcoded for $n\leq 31$ [18], so it has no configuration at this population and cannot be run without hand-designing new ones—a concrete scalability ceiling in the method, not a tuning choice. At this scale each client holds only $\sim$100 samples and attacker exclusion consumes the first $\sim$15–20 rounds, so a short 30-round budget truncates every method mid-recovery; extending to 50 rounds lets recovery complete and collapses

TABLE III
SCALING TO N=500 CLIENTS (FASHIONMNIST, SIGN-FLIP f=0.2, 50 ROUNDS, DIRICHLET α=0.5). TEST ACCURACY IS THE MEAN OF THE FINAL FIVE ROUNDS OVER SEEDS $\{0, 2, 3\}$ ($\pm$1 SD); HONEST FP IS THE NUMBER OF HONEST CLIENTS (OF 400) WRONGLY EXCLUDED. FEDGT HAS NO CONFIGURATION AT THIS SCALE—ITS PARITY-CHECK DESIGN IS HARDCODED FOR $n\leq 31$ [18], A CONCRETE SCALABILITY CEILING. ONLY CB-SAFE+ HOLDS NEAR-CLEAN ACCURACY WHILE COORDINATE-WISE RULES COLLAPSE, WITHOUT DISCARDING HONEST CLIENTS AND WITH POST-QUANTUM SECURITY.

Rule	Acc. (%)	Honest FP↓	PQC
CB-SAFE+ (ours)	**65.7** $\pm$ 2.3	11	✓
FedGT [18]	– (no config, $n\leq 31$)	–	–
FLTrust [10]	61.3 $\pm$ 2.4	–	–
Median	25.1 $\pm$ 7.5	–	–
Geo-median/RFA [20]	28.9 $\pm$ 9.9	–	–
Multi-Krum	10.0 $\pm$ 0.1	–	–
Bulyan [19]	10.8 $\pm$ 1.3	–	–
Trimmed mean	9.5 $\pm$ 0.9	–	–
Mean	10.0 $\pm$ 0.0	–	–

the between-seed spread (from ± 10 to ± 2, Fig. 3). At the converged horizon CB-SAFE+ reaches 65.7% (last-five-round mean over three seeds), the coordinate-wise rules remain laundered near chance, and FLTrust's trust weighting, which recovers fastest early, plateaus $\sim$4 points lower (61.3%) because soft down-weighting never fully removes the laundered mass. CB-SAFE+ reaches that accuracy while excluding only 11 honest clients of 400 (2.75%). Among the methods that admit a configuration at this scale, CB-SAFE+ is the only one that pairs top-tier robustness with near-zero honest exclusion and post-quantum security.

D. Utility and Cryptographic Equivalence

Plain FedAvg reaches 59.0% test accuracy (mean of the final five of 40 rounds; 59.6% at round 40) on the non-IID split. In every round, the same client updates were pushed through the full cryptographic pipeline twice (once under HQC-128, once under ML-KEM-768), and every per-cluster secure sum was compared coordinate-wise against the plain sum. The maximum deviation across all 40 rounds and both KEMs was 2.28×10^{-5}, which *equals* the worst-case fixed-point rounding bound for a 3-element sum ($3\cdot 2^{-17} = 2.29\times 10^{-5}$): the masking arithmetic itself is exact, so CB-SAFE's accuracy curve is identical to plain FedAvg by construction, not approximately. Both dropout modes were exercised: in round 10, two of three dropped clients landed in the same cluster, driving it below its t=2 threshold: the protocol excluded that cluster and recovered the remaining nine exactly; in round 20, three dropouts in three distinct clusters were all recovered by seed-reveal plus Shamir reconstruction. Measured in-run, setup cost 2.6 s (HQC-128) vs. 0.02 s (ML-KEM-768) for all 30 clients, and steady-state rounds took $\sim$30 ms of client-side masking and $\sim$0.29 s of server-side unmasking, negligible against the $\sim$21 s of local training per round.

E. Robustness Under Attack

The contamination law holds. Across all runs, the realized fraction of clean cluster sums tracked $(1-f)^c$ closely: at c=3,

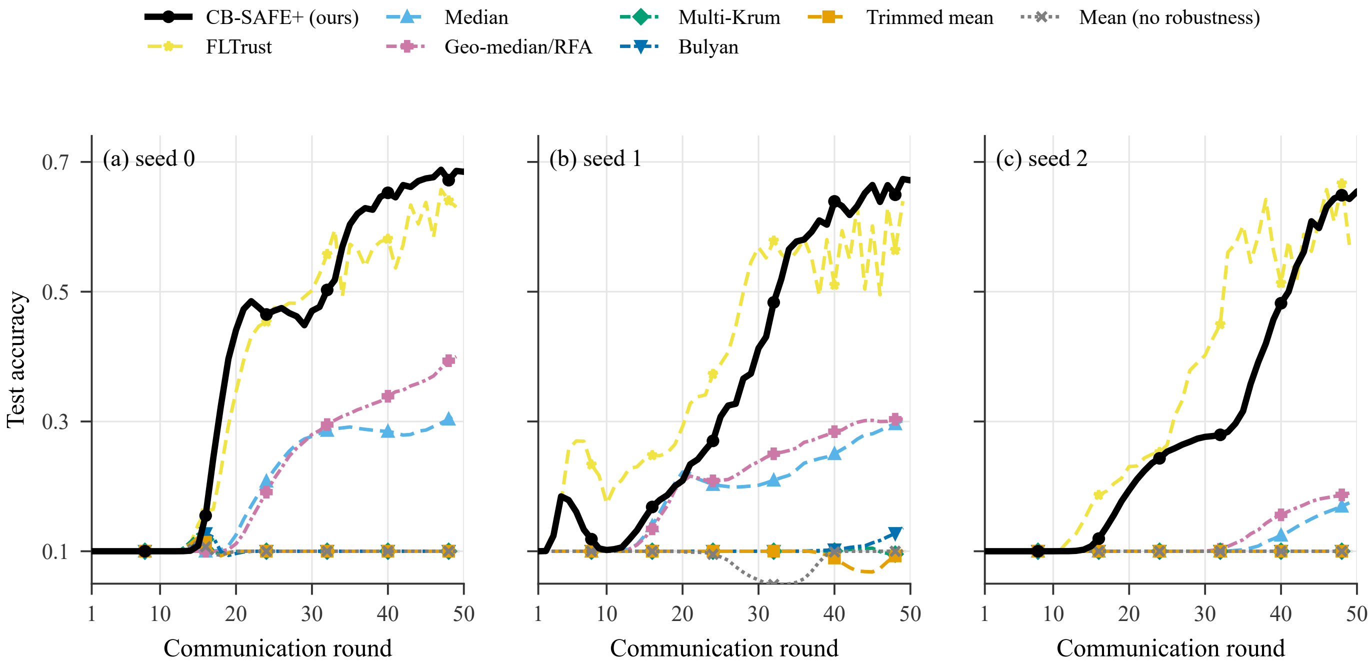

Fig. 3. Test accuracy vs round at N=500 (FashionMNIST, sign-flip f=0.2), one panel per seed; eight rules (FedGT excluded, see text).

TABLE IV
ROBUSTNESS OVER CLUSTER SUMS (c=3, k=10; 3 SEEDS, 50 ROUNDS): LAST-FIVE-ROUND ACCURACY UNDER SIGN-FLIP AND BACKDOOR ASR (LOWER BETTER) VS MALICIOUS FRACTION f. BEST IN BOLD.

	Sign-flip acc. (%)			Backdoor ASR (%)		
Rule	f=.1	f=.2	f=.3	f=.1	f=.2	f=.3
Mean	20.0	10.0	10.0	89.8	95.6	97.0
Trimmed mean	51.7	10.0	10.0	85.7	95.1	96.7
Median	55.5	10.0	10.0	66.9	94.8	96.7
Multi-Krum	**60.5**	45.3	21.4	**63.7**	94.8	97.0
CB-SAFE+	60.5	**57.5**	**57.4**	88.1	95.1	97.0

observed 0.70/0.50/0.40 versus predicted 0.73/0.51/0.34 for f=0.1/0.2/0.3.

Sign-flip: laundering collapses coordinate-wise rules exactly at the predicted boundary (Table IV, Fig. 6). At f=0.1 (≈3 dirty clusters, within trim capacity) the static rules work: trimmed mean 51.7%, median 55.5%, Multi-Krum 60.5%, versus 20.0% for the undefended mean and ∼60% clean baseline. At f=0.2 (≈5 dirty clusters) every coordinate-wise rule is pinned at exactly 10.0% (random guessing), not because the model diverges but because the laundered $\pm h$ population drives the aggregate to zero (Proposition 1), while Multi-Krum, which selects whole vectors rather than per-coordinate statistics, still reaches 45.3% (21.4% at f=0.3). Robust rules that rely on magnitude order statistics are structurally weakened by the very mechanism that provides privacy; vector-selection rules retain power.

The collapse is a wide basin, not a knife-edge. Since γ=5 is the exact laundering point γ^*=$2c/m-1$, one might ask whether the collapse is an artifact of that singular value. Sweeping $\gamma \in \{2,3,5,8,15\}$ at f=0.2 (Fig. 4) shows it is not: median and trimmed mean collapse to 10% at *every* $\gamma \geq \gamma^*$

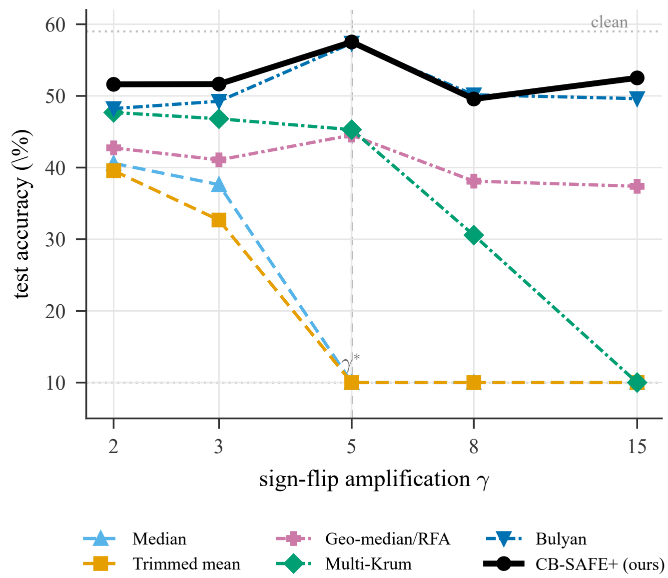

Fig. 4. Sign-flip amplification sweep (CIFAR-10, f=0.2, 3 seeds): test accuracy vs γ. Coordinate-wise rules collapse for all $\gamma \geq \gamma^*$=5; dashed lines mark chance and the clean baseline.

and only partially degrade below it, and Multi-Krum—robust until γ=8—itself collapses at γ=15. CB-SAFE+ holds ∼50–53% throughout. Bulyan matches it here, its trim capacity not exceeded at f=0.2; that parity breaks at f=0.3 (Table VI), where identifying attackers rather than merely trimming them decides the outcome.

The full baseline panel confirms the mechanism (Fig. 6, Table VI). Averaged over three seeds, the eight rules separate into three regimes at f=0.3: the coordinate-wise rules (mean, trimmed mean, median) are laundered to chance (≈10%) and

TABLE V
ATTACKER IDENTIFICATION, CIFAR-10 SIGN-FLIP (N=30, 3 SEEDS): ATTACKERS CAUGHT (↑) AND HONEST FALSE POSITIVES (↓). FEDGT IS ITS REAL BCJR DECODER [18]. BEST IN BOLD.

	f=0.1		f=0.2		f=0.3	
Method	caught↑	FP↓	caught↑	FP↓	caught↑	FP↓
FedGT [18]	2.7/3	13.0	5.3/6	17.7	7.7/9	11.0
CB-SAFE+ (ov1)	3/3	0.7	4/6	0.7	6/9	1.3
CB-SAFE+ hybrid	**3/3**	**0.3**	**6/6**	**0.3**	**9/9**	**0.0**

the robust geometric median RFA [20] degrades to ∼20%, confirming that location- and magnitude-based aggregation fails regardless of sophistication; the vector-selection family degrades but falls (Multi-Krum 21%, Bulyan [19] 35%); and only CB-SAFE+ holds near the clean baseline at 57.4%, whereas FedGT's actual BCJR decoder [18], run per round, degrades to 34% on CIFAR-10 (and collapses to near-chance on FashionMNIST and EMNIST) at this contamination by over-excluding honest clients (Table V). That RFA collapses while Bulyan merely degrades sharpens Proposition 1: laundering specifically defeats rules that trust the *aggregate* direction or magnitude, precisely what cluster averaging preserves for the attacker, whereas whole-vector selection resists longer.

CB-SAFE+ restores learning past the static breakdown (Table IV). With loss-probe flags and temporal suspicion, the defense identifies all 3 of 3 attackers at f=0.1 (60.5%, at the clean ceiling) and all 9 sign-flip attackers with *zero* honest false positives at f=0.3 (three-seed means in Table V), recovering to 57.4% accuracy where every static coordinate-wise rule sat at 10% and Multi-Krum at 21.4%—exceeding even the 39–41% the median achieves *without any privacy* (c=1), i.e., the defense refunds the privacy dial's cost in rounds rather than anonymity. At f=0.2 it holds 57.5% while Multi-Krum reaches only 45.3%, so the defense's advantage over vector-selection rules grows with attack severity. Under label flipping (a weak attack in this regime: all rules land within 44–52%) the probe flags only the worst clusters and excludes conservatively (zero false positives at f=0.2), matching the best static rule.

Backdoor: a mechanism-independent blind spot. The trigger attack leaves main-task accuracy intact for every rule (50–55%) while ASR stays at 86–97% across mean, trimmed mean, Multi-Krum, *and* CB-SAFE+ (Fig. 5): backdoored cluster sums *reduce* clean root loss and align with the honest direction, so they are invisible to loss- and direction-based tests alike. The one partial exception is the coordinate-wise median at f=0.1, which suppresses ASR to 67% on CIFAR-10 (and to 42% on FashionMNIST): order statistics can filter the trigger direction's coordinates when contamination is sparse. We therefore claim CB-SAFE+ for *untargeted* model poisoning only; targeted trigger attacks under secure aggregation remain open, consistent with their difficulty even without privacy constraints.

F. Generalization Across Datasets and Modalities

To test whether the laundering effect and CB-SAFE+'s recovery are artifacts of one dataset, we repeat the sign-flip

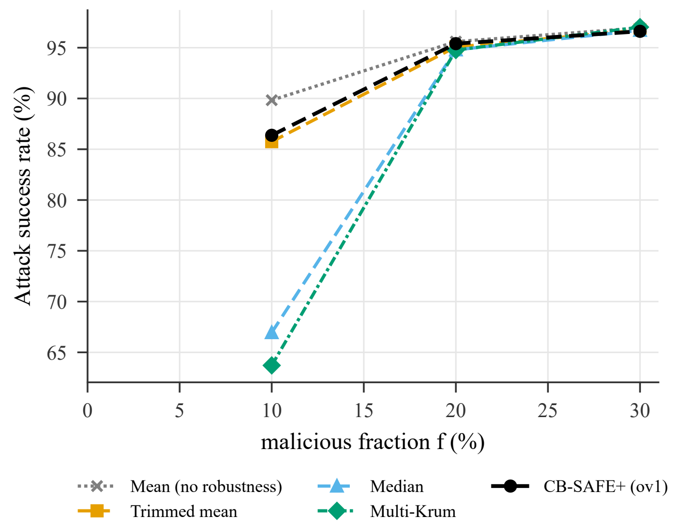

Fig. 5. Backdoor attack success rate vs. malicious fraction (CIFAR-10); lower is better.

evaluation on FashionMNIST (a second image benchmark, three seeds) and on Edge-IIoTset (tabular IoT/IIoT intrusion detection with a multilayer perceptron, a different modality in the security domain of the closest prior work [7]). The pattern is identical everywhere (Table VI, Fig. 6): every coordinate-wise rule collapses to chance at $f{\geq}0.2$ (10.0% on the 10-class image sets, 7.3% on 15-class Edge-IIoTset), Multi-Krum degrades gracefully then falls at f=0.3, and CB-SAFE+ tracks the clean baseline. The round-by-round view (Fig. 6) shows this is not an endpoint artifact: the static rules decay progressively while CB-SAFE+ dips during warmup and recovers as exclusions fire. At f=0.3 CB-SAFE+ retains ∼95%, ∼99%, and ∼99% of the no-attack accuracy on CIFAR-10, FashionMNIST, and Edge-IIoTset respectively, versus 12–18% for the worst static rule. Pooling all matched (dataset, f, seed) cells, CB-SAFE+ beats every baseline under a paired t-test with Holm correction: versus the mean by 57.3 points ($p{<}10^{-15}$), the median by 44.0 ($p{<}10^{-8}$), Multi-Krum by 18.6 ($p{<}10^{-3}$), and FedGT's actual BCJR decoder—over the three datasets where FedGT admits a configuration—by 21.0 ($p{<}10^{-3}$). The Edge-IIoTset result is the sharpest: at f=0.3 CB-SAFE+ holds 64.5% against a ∼65% clean baseline while mean, trimmed mean, and median all sit at 7.4%.

G. Secondary Attack: Label Flipping

Label flipping is a far milder attack than sign-flip, and it doubles as a control. Table VII shows that *no* coordinate-wise rule collapses: the undefended mean retains 53–64% at f=0.3, versus 10% under sign-flip. That contrast isolates the laundering mechanism (Proposition 1) as the cause of the sign-flip failures, rather than any generic fragility of the setup. CB-SAFE+ stays at or above every baseline, and on FashionMNIST it holds 87% at f=0.3 where the mean falls to 59%. FedGT's actual BCJR decoder survives this mild attack—unlike its sign-flip collapse—but trails CB-SAFE+ on CIFAR-10 (∼50–56% vs ∼56–61%) and is noisier between

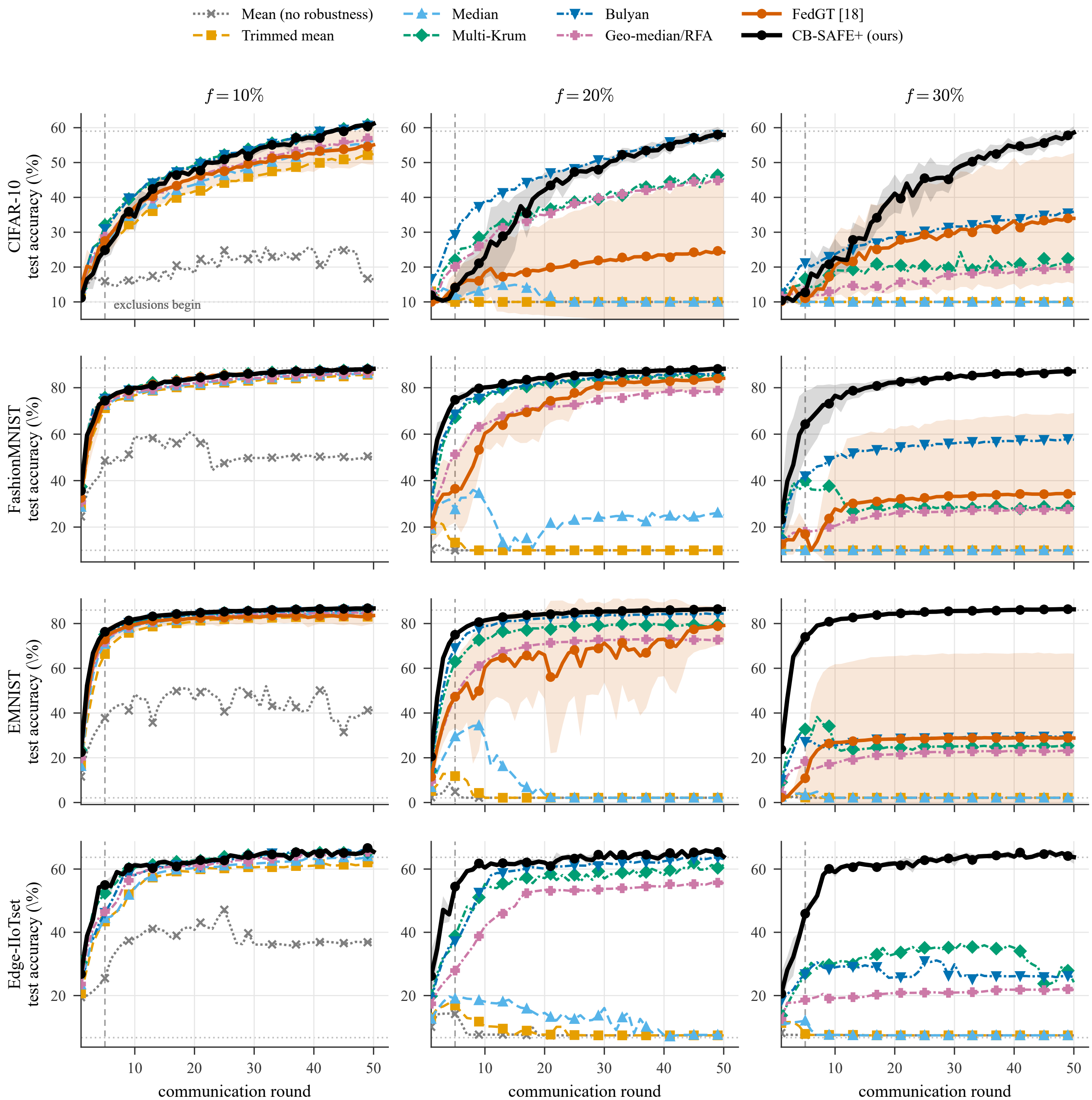

Fig. 6. Training dynamics under sign-flip: test accuracy vs round, eight rules across four datasets (rows) $\times$ malicious fraction f (columns). Mean over 3 seeds; ± 1 SD bands for CB-SAFE+ and FedGT (real BCJR [18], no Edge-IIoTset config). Dashed line: warmup $W = 5$.

TABLE VI

TEST ACCURACY (%) UNDER THE SIGN-FLIP (LAUNDERING) ATTACK ACROSS FOUR DATASETS AND MALICIOUS FRACTIONS f (MEAN ± STD OVER 3 SEEDS, 50 ROUNDS). HIGHER IS BETTER; BEST PER COLUMN IN BOLD. RED ENTRIES ARE PROVISIONAL (RUNS STILL COMPLETING). A DASH (–) MARKS A CONFIGURATION NOT YET EVALUATED. CB-SAFE+ VARIANTS ARE ABLATED IN TABLE IX.

	CIFAR-10			FashionMNIST			EMNIST			Edge-IIoTset		
Method	f=.1	f=.2	f=.3	f=.1	f=.2	f=.3	f=.1	f=.2	f=.3	f=.1	f=.2	f=.3
FedAvg (mean)	20.0 ±8.3	10.0 ±0.0	10.0 ±0.0	49.7 ±29.2	10.0 ±0.0	10.0 ±0.0	39.7 ±26.7	2.1 ±0.0	2.1 ±0.0	36.8 ±21.7	7.4 ±0.2	7.4 ±0.2
Trimmed mean	51.7 ±1.0	10.0 ±0.0	10.0 ±0.0	85.5 ±1.0	10.0 ±0.0	10.0 ±0.0	83.1 ±0.3	2.1 ±0.0	2.1 ±0.0	61.7 ±2.6	7.4 ±0.2	7.4 ±0.2
Median	55.5 ±1.5	10.0 ±0.0	10.0 ±0.0	86.0 ±0.7	25.4 ±21.8	10.0 ±0.0	84.6 ±0.0	2.1 ±0.0	2.1 ±0.0	63.7 ±0.6	7.6 ±0.2	7.4 ±0.2
Multi-Krum	**60.5 ±0.6**	45.3 ±2.0	21.4 ±11.6	87.6 ±0.4	85.0 ±1.2	28.1 ±25.5	86.0 ±0.2	79.2 ±0.8	25.2 ±32.7	65.2 ±0.2	60.5 ±1.3	25.9 ±23.2
Bulyan	60.3 ±0.5	57.4 ±0.6	35.4 ±19.4	87.3 ±0.3	86.0 ±1.0	57.6 ±33.7	85.8 ±0.3	84.4 ±0.3	29.5 ±38.7	**65.5 ±0.4**	63.3 ±0.9	26.0 ±26.2
Geo-median / RFA	56.5 ±0.7	44.5 ±1.3	19.5 ±13.5	86.3 ±0.7	78.6 ±2.1	27.5 ±24.8	84.9 ±0.2	72.8 ±1.4	23.0 ±29.5	65.1 ±0.4	55.4 ±9.2	21.9 ±20.4
FLTrust	34.6 ±5.7	34.2 ±5.0	34.2 ±6.4	79.1 ±0.1	79.9 ±1.2	78.6 ±0.6	76.3 ±0.8	75.7 ±1.2	75.0 ±1.3	50.4 ±3.7	43.3 ±6.7	39.9 ±1.4
FedGT [TIFS'25]	54.4 ±5.2	24.2 ±20.1	33.7 ±18.3	87.9 ±0.3	83.7 ±4.7	34.3 ±34.4	83.4 ±3.9	78.6 ±9.1	28.8 ±37.8	–	–	–
CB-SAFE+ (ours)	60.5 ±0.5	**57.5 ±1.6**	**57.4 ±1.8**	**87.9 ±0.3**	**88.0 ±0.4**	**86.9 ±0.8**	**86.7 ±0.2**	**86.4 ±0.1**	**86.4 ±0.2**	65.4 ±0.5	**65.2 ±0.2**	**64.5 ±0.8**

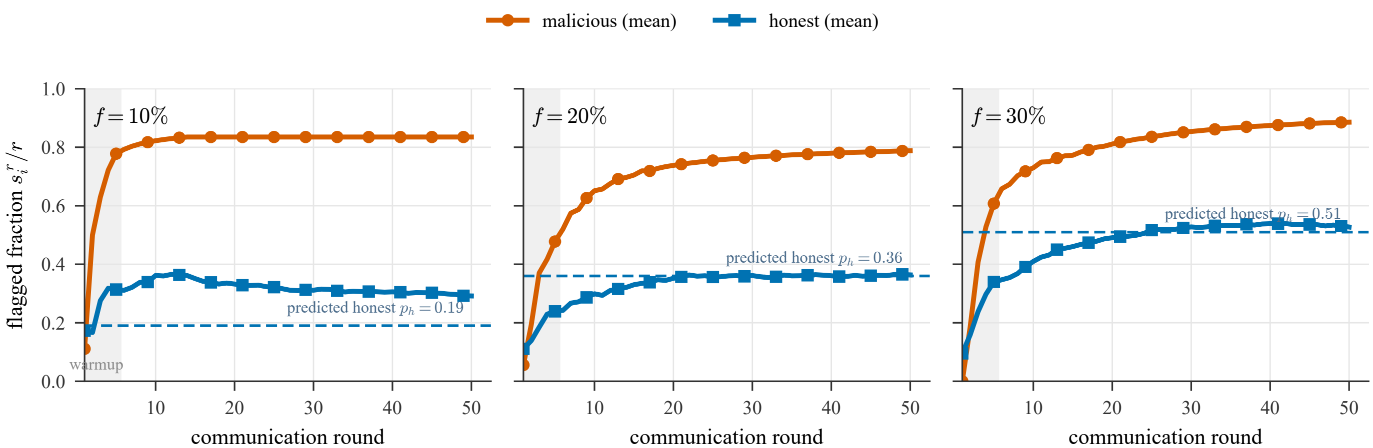

Fig. 7. Temporal suspicion (CB-SAFE+, CIFAR-10): mean flagged fraction s_i^r/r for malicious vs honest clients, per fraction f (3 seeds). Dashed: predicted honest rate p_h=$1-(1-f)^{c-1}$; warmup shaded.

seeds, so even where its accuracy signal is informative it is the less stable identifier.

H. Adaptive Attacker: Duty-Cycled Poisoning

The exclusion rule $s_i^r/r > \tau$ (τ=0.85) invites an obvious evasion: poison only a fraction $\delta < \tau$ of rounds, keeping most of the attack while never crossing the threshold. We test this directly (Table VIII), and it fails. Because the partition is re-randomized every round, a resting attacker is still flagged whenever it co-clusters with an *active* one, so its accumulated suspicion runs ahead of its nominal duty: at δ=0.84, right at the threshold, CB-SAFE+ still excludes all six attackers in every seed. Across $\delta \in [0.5, 1.0]$ it excludes 4–6 of 6 and accuracy never drops below ∼50% (clean baseline 59%), where coordinate-wise rules sit at 10%. The duty-cycler faces a no-win trade-off: raise δ and it is caught; lower it and the attack is correspondingly weaker (accuracy rises to 53.7%). A stronger threshold-aware adversary that jointly minimizes footprint and detectability remains open, but the naive duty-cycle evasion does not succeed.

I. Ablation: Which Component Earns the Robustness

CB-SAFE+ layers three ingredients on top of clustered secure aggregation: temporal reputation, overlapping re-randomized groups, and a per-round COMP decode. Table IX isolates each under the identical sign-flip protocol. The base temporal reputation (ov1) already restores learning at moderate contamination but degrades at f=0.3 (37.9% on CIFAR-10); adding overlapping groups (ov4) lifts it to 48.8%, and the full hybrid COMP decode reaches 50.0%— far above FedGT's actual BCJR decoder, which collapses to chance at this contamination (Table V). Each component contributes, and the gains concentrate in the high-f regime where laundering is strongest. The trust-free variant, which uses *no* root dataset, matches the others at f=0.1 but collapses beyond it: attacker identification without any trusted anchor is possible only in the low-contamination regime, a capability boundary we state rather than hide.

J. The Privacy–Robustness Dial, Measured

Fixing the rule (median) and sweeping cluster size traces the dial empirically (Table Xa, Fig. 8). Without privacy (c=1), the median degrades gently from 50% to 39% as f grows to 0.3, i.e., classical robust aggregation. At c=3 the same rule holds to f=0.1 (43.4%) and collapses beyond; at c=5 degradation begins already at f=0.1 (33.7%): the breakdown frontier shifts left exactly as $(1-f)^c$ predicts (Fig. 8). Two further observations sharpen the picture. First, at f=0.05 privacy is *free*: c=3 and c=5 match or slightly exceed c=1

TABLE VII

TEST ACCURACY (%) UNDER THE LABEL-FLIP ATTACK ACROSS FOUR DATASETS AND MALICIOUS FRACTIONS f (MEAN ± STD OVER 3 SEEDS, 50 ROUNDS). LABEL FLIPPING IS A MILD ATTACK: UNLIKE SIGN-FLIP (TABLE VI), COORDINATE-WISE RULES DO *not* COLLAPSE, WHICH ISOLATES LAUNDERING AS THE MECHANISM BEHIND THE SIGN-FLIP FAILURES. HIGHER IS BETTER; BEST PER COLUMN IN BOLD. RED ENTRIES ARE PROVISIONAL. A DASH (–) MARKS A CONFIGURATION NOT EVALUATED.

	CIFAR-10			FashionMNIST			EMNIST			Edge-IIoTset		
Method	f=.1	f=.2	f=.3	f=.1	f=.2	f=.3	f=.1	f=.2	f=.3	f=.1	f=.2	f=.3
FedAvg (mean)	60.4 ±0.6	57.1 ±1.0	53.1 ±1.5	87.2 ±0.5	80.0 ±0.4	58.9 ±5.2	85.6 ±0.0	82.8 ±0.1	64.1 ±2.0	64.7 ±0.3	65.5 ±1.0	64.1 ±0.7
Trimmed mean	60.1 ±0.3	57.2 ±0.2	54.2 ±1.1	87.3 ±0.4	82.8 ±0.3	72.1 ±3.0	86.1 ±0.2	83.9 ±0.1	79.1 ±1.3	65.3 ±0.9	**66.3 ±1.2**	64.6 ±0.8
Median	60.1 ±0.4	57.3 ±0.1	53.6 ±2.0	87.4 ±0.3	86.0 ±0.5	74.4 ±5.5	86.2 ±0.2	84.0 ±0.2	79.9 ±1.5	65.0 ±1.2	65.6 ±0.6	64.7 ±1.1
Multi-Krum	59.8 ±0.2	58.1 ±1.3	55.1 ±4.4	87.6 ±0.3	87.9 ±0.3	80.1 ±7.9	85.9 ±0.2	84.9 ±0.1	81.4 ±3.3	64.8 ±0.4	65.1 ±0.7	64.5 ±0.7
Bulyan	58.8 ±0.5	57.0 ±1.3	54.5 ±3.4	87.3 ±0.3	87.2 ±0.4	77.4 ±10.0	85.8 ±0.3	84.5 ±0.3	81.8 ±2.9	64.6 ±0.2	63.9 ±0.7	63.1 ±0.4
Geo-median / RFA	**60.7 ±0.4**	57.7 ±1.0	54.5 ±1.8	87.6 ±0.4	86.5 ±0.7	72.0 ±6.3	86.3 ±0.1	83.9 ±0.3	79.6 ±1.3	**65.3 ±1.3**	65.8 ±0.7	**64.8 ±0.3**
FLTrust	37.6 ±3.3	36.1 ±3.3	35.6 ±4.0	79.4 ±0.1	80.4 ±0.4	78.8 ±0.8	76.4 ±0.9	75.7 ±1.0	75.2 ±1.4	50.5 ±3.1	44.0 ±5.0	43.1 ±4.0
FedGT [TIFS'25]	55.8 ±2.1	50.2 ±8.2	50.3 ±5.9	87.7 ±0.4	87.7 ±0.7	82.1 ±2.3	85.7 ±0.0	84.9 ±0.6	80.7 ±1.3	–	–	–
CB-SAFE+ (ours)	60.6 ±0.3	**59.0 ±0.3**	**56.5 ±1.3**	**87.9 ±0.4**	**88.0 ±0.3**	**87.0 ±0.8**	**86.7 ±0.1**	**86.3 ±0.2**	**86.4 ±0.2**	65.3 ±0.5	64.3 ±0.2	63.3 ±0.9

TABLE VIII

DUTY-CYCLED ATTACKER (CIFAR-10 SIGN-FLIP, γ=5, f=0.2; 3 SEEDS, 50 ROUNDS): A CLIENT POISONS A FRACTION δ OF ROUNDS (δ=1: ALWAYS ON); EXCLUSION THRESHOLD τ=0.85.

Attacker duty δ	Test acc. (%)	Attackers excl. (of 6)
Clean (no attack)	59.0	–
0.50	59.8 ±0.5	1.7
0.70	59.0 ±0.5	5.3
0.80	59.1 ±1.0	6.0
0.84	57.7 ±1.6	6.0
0.90	57.1 ±2.4	5.7
1.00 (always-on)	57.5 ±1.6	6.0

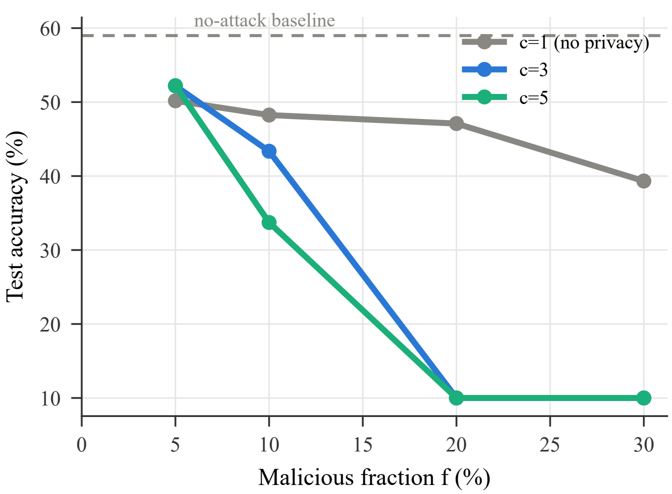

Fig. 8. Privacy–robustness dial: median-rule accuracy under sign-flip for $c \in \{1, 3, 5\}$ (CIFAR-10).

(52.2% vs. 50.2%), because cluster averaging denoises before the median. Second, the dial's cost is refundable: at c=3, f=0.3, where the private median collapses to 10% against 39–41% without privacy, CB-SAFE+ recovers comparable accuracy *while keeping the anonymity set*, by paying in rounds (warmup) rather than in privacy.

VI. SECURITY DISCUSSION AND LIMITATIONS

What the design guarantees. Within a cluster, privacy reduces to the double-masking argument of [1] with the DH-style key agreement replaced by an IND-CCA2 KEM: masks are ChaCha20 outputs keyed by HKDF of encapsulated secrets, so an honest-but-curious server colluding with fewer than t cluster members learns only cluster sums. The robust layer never inspects anything finer than a cluster mean. **What it does not.** We do not claim security against an actively malicious server (who could misreport dropouts; standard mitigations apply), nor differential privacy on the cluster sums themselves; the anonymity set is bounded by the alive cluster size, and cluster sums of very small clusters leak more than full-cohort sums. Robustness degrades predictably as $k(1-f)^c$ approaches the trim capacity: at c=3, f=0.3 leaves an expected 3.4 clean clusters of 10, near the breakdown of a 2-per-side trimmed mean; this is the price of the anonymity set and we report it rather than hide it. HQC operations are $\sim$200$\times$ slower than ML-KEM's, which is immaterial here (setup-only) but would matter for protocols that encapsulate per round.

VII. CONCLUSION

CB-SAFE positions cryptographic diversity as a configuration choice rather than a research programme: the KEM sits behind one interface, and the measured cost of choosing the code-based occupant is a one-time setup constant. The more consequential finding is what secure aggregation does to robustness: cluster averaging does not merely dilute poisoning, it *launders* it, transforming conspicuous attacks into plausible-looking sums that defeat exactly the rules practitioners would reach for. Defending therefore requires signals that survive aggregation (direction, loss) accumulated over time. CB-SAFE+ shows one such design that identifies individual attackers from group observations without spending any privacy. The open questions our results make concrete: targeted backdoors remain invisible to sum-level probes at every malicious fraction we tested, an actively malicious server is outside our threat model, and while a naive duty-cycled attacker does not evade the temporal test (Table VIII), a threshold-aware adversary that jointly minimizes its footprint and detectability remains the natural next step. The implementation and all measurement scripts are released open-source as, to our knowledge, the first code-based reference point for post-quantum secure aggregation in FL.

TABLE IX
ABLATION OF CB-SAFE+ COMPONENTS UNDER SIGN-FLIP (SAME PROTOCOL AS TABLE VI). OVERLAPPING GROUPS AND THE HYBRID COMP DECODE EACH ADD ROBUSTNESS AT HIGH f; THE TRUST-FREE VARIANT USES NO ROOT DATASET BUT DEGRADES BEYOND f=0.1. BEST PER COLUMN IN BOLD.

Method	CIFAR-10			FashionMNIST			EMNIST			Edge-IIoTset		
	f=.1	f=.2	f=.3	f=.1	f=.2	f=.3	f=.1	f=.2	f=.3	f=.1	f=.2	f=.3
Base: temporal reputation (ov1)	61.1 ±0.3	55.9 ±1.8	53.6 ±5.6	**88.0 ±0.4**	**88.0 ±0.3**	86.7 ±1.0	86.7 ±0.2	86.0 ±0.2	86.3 ±0.2	65.3 ±0.4	65.1 ±1.1	**64.5 ±0.7**
+ overlapping groups (ov4)	60.4 ±0.6	**57.6 ±1.6**	**57.7 ±2.3**	87.9 ±0.4	88.0 ±0.3	**86.9 ±0.8**	86.7 ±0.1	86.3 ±0.1	**86.5 ±0.2**	**65.4 ±0.5**	**65.2 ±0.2**	64.5 ±0.8
+ hybrid COMP decode (full)	60.5 ±0.5	57.5 ±1.6	57.4 ±1.8	87.9 ±0.3	88.0 ±0.4	86.9 ±0.8	86.7 ±0.2	**86.4 ±0.1**	86.4 ±0.2	65.4 ±0.5	65.2 ±0.2	64.5 ±0.8
Trust-free (no root data)	**61.4 ±0.4**	10.0 ±0.0	10.0 ±0.0	87.8 ±0.3	10.0 ±0.0	10.0 ±0.0	**86.8 ±0.1**	2.1 ±0.0	2.1 ±0.0	64.4 ±0.3	45.2 ±26.6	7.4 ±0.2

TABLE X
ABLATIONS ON CIFAR-10 SIGN-FLIP (FINAL ACCURACY, %). (A) THE CLUSTER-SIZE DIAL (MEDIAN RULE). (B) OVERLAP DEGREE o (CB-SAFE+ REPUTATION).

(a) cluster size c (rule: median)

	f=.05	f=.1	f=.2	f=.3
c=1 (no privacy)	51	49	45	40
c=3	53	49	10	10
c=5	52	40	10	10

(b) overlap degree o (rule: CB-SAFE+ reputation)

	f=.1	f=.2	f=.3
o=1	54	48	39
o=4	54	51	49

APPENDIX A
DATASET DETAILS AND MODELS

All datasets are partitioned across N=30 clients by a Dirichlet(α=0.5) label-skew split [47]; each client runs one local epoch of SGD per round. **CIFAR-10** [43]: 50,000 training / 10,000 test images, 10 classes; a convolutional network with d=291,498 parameters (lr 0.01, momentum 0.9, batch 64). **FashionMNIST** [44]: 60,000/10,000 grayscale images, 10 classes, same network adapted to one input channel. **EMNIST-balanced** [45]: 112,800/18,800 handwritten characters, 47 classes, a harder label space; same convolutional backbone. **Edge-IIoTset** [46]: a tabular IoT/IIoT intrusion-detection corpus with 15 classes, class-stratified subsampling to at most 120,000 rows, leakage/identifier columns dropped, features standardized, classified by a multilayer perceptron. Image datasets use 30 rounds; Edge-IIoTset uses 25.

APPENDIX B
ATTACK CONFIGURATIONS

All attacks are offline (training-time) data/model poisoning at malicious fractions $f \in \{0.1, 0.2, 0.3\}$. **Sign-flip:** each malicious client submits $-\gamma$ times its honest update with amplification γ=5, the setting at which laundering is exact for c=3, m=1 (Proposition 1). **Label-flip:** labels are deterministically remapped ($y \to 9-y$ on the 10-class image sets), a comparatively weak attack in this regime. **Backdoor:** a 3×3 trigger patch with target class 0 and a 50% local poison rate [15], evaluated by attack success rate (ASR).

APPENDIX C
EXTENDED HYPERPARAMETERS AND REPRODUCIBILITY

Unless stated: cluster size c=3 (k=10 clusters), Shamir threshold t=$\lfloor c/2 \rfloor$+1=2, robust rule trimmed mean/median with trim β per side, overlap degree $o \in \{1, 4\}$, server root $|\mathcal{D}_0|$=200 held out from all clients, exclusion threshold τ=0.85 (or the adaptive gap rule) after a warmup of W=5 rounds. KEMs span HQC-128/192/256 and ML-KEM-512/768/1024 at matched NIST levels; masks are ChaCha20 keystreams with HKDF-SHA256 per-round seeds. Runs use Python 3.12, PyTorch 2.5.0 (CUDA 12.4), liboqs-python 0.15.0 on a single NVIDIA RTX 2060 (6 GB). CIFAR-10, FashionMNIST, and Edge-IIoTset results are averaged over three seeds and EMNIST over one; significance is assessed by a paired t-test with Holm correction. Every run emits a per-round CSV and is reproducible from the released scripts.

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